
CF11: COMPLETE FORMALISM — DARK SECTOR EMERGENCE FROM METRIPLECTIC DYNAMICS IN VDM

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Abstract

This Complete Formalism (CF11) specifies a falsifiable, derived-limit route by which the VDM metriplectic split (reversible J limb $\oplus$ irreversible M limb) can be interpreted as an *effective* two-component “dark sector” at cosmological scales without postulating new fundamental particles. The central technical move is to avoid an ad hoc field split $\Phi = \Phi_J + \Phi_M$: instead, a state-level decomposition is defined by spectral (Riesz) projectors of the linearized generator about a coarse-grained background, producing oscillatory (reversible) and relaxational (irreversible) bands with a measurable spectral-gap gate. Sector stress–energy and a tracked interaction remainder are then defined in a quadratic approximation using the energy Hessian inner product, yielding explicit cross-term control metrics. Within clearly stated regime diagnostics, CF11 derives correct bounded equation-of-state limits: a coherent, slowly varying, nearly homogeneous sector satisfies $w_J \approx -1$, while a defect/oscillation-dominated, non-relativistic sector admits $w_M \approx 0$ via either particle-kinematics or rapid oscillations in a quadratic minimum. The deliverable is a publishable contract plus binary gates for (i) T2 lattice validation of degeneracy, entropy monotonicity, the spectral split, and causality bounds, and (ii) T3+ observational regression using model selection (AIC/BIC) and a hierarchy–lensing bias statistic.

Keywords dark sector · interacting fluids · equation of state · metriplectic dynamics · GENERIC · Riesz projection · spectral gap · scalar field cosmology · telegraph equation · causality · hierarchical interfaces · Void Dynamics Model

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1 Objective, scope, and non-claims

VDM uses a metriplectic evolution law (reversible J limb $\oplus$ irreversible M limb) as an axiom-core contract on admissible dynamics [Lietz, 2025a, Morrison, 1986, Grmela and Öttinger, 1997, Öttinger and Grmela, 1997]. CF11 is *derived-limit*: it proposes a cosmology-facing *interpretation layer* built on the metriplectic split, then reduces that interpretation to measurable gates that can be validated in lattice simulations (T2) and challenged on cosmological datasets (T3+).

One-line objective. To define a mathematically standard, numerically implementable decomposition of coarse-grained dynamics into reversible/oscillatory and irreversible/relaxational sectors, and to map that decomposition to a two-component effective “dark sector” bookkeeping with falsifiable gates.

What CF11 does.

1. Provides a **well-posed decomposition of motion** into J and M channels (imported from CF01/CF02) together with an explicit *state-level* split defined by spectral projectors of a linearized generator [Lietz, 2026a].
2. Defines an **operational split of coarse-grained stress–energy** into two effective components J and M , plus a tracked interaction remainder, using mode projections rather than an ad hoc field split.
3. Derives **equation-of-state (EoS) limits** in controlled regimes: $w_J \approx -1$ in a coherent, slowly varying, nearly homogeneous regime; and $w_M \approx 0$ in either a defect-gas regime or a rapid-oscillation regime near a quadratic minimum [Mukhanov, 2005, Turner, 1983].
4. Exposes the interpretation as **binary falsification gates**: (i) internal simulation gates (degeneracy, entropy, spectral gap, cross-term control, causality); and (ii) observational regression gates (model selection via AIC/BIC; hierarchy–lensing bias).

What CF11 does not claim.

- **No new fundamental particles.** “Dark sector” is an effective bookkeeping interpretation of coarse-grained dynamics, not a postulate of new microscopic degrees of freedom.
- **No globally unique field identity.** CF11 does not assume a globally unique split $\Phi = \Phi_J + \Phi_M$ for all states; the split is defined only after a declared projection choice (Section 5.2).
- **No covariant dissipation-as-Lagrangian.** Dissipation enters through M and Σ consistent with metriplectic/GENERIC structure, not by smuggling friction into a covariant action [Öttinger, 2005].
- **No “cross-term cancellation” without measurement.** Interaction remainders are measured and gated; they are not assumed small by fiat (Section 5.3).

Prerequisites (internal). CF11 imports contracts and definitions from earlier Complete Formalisms: CF01 (QGT $\rightarrow J \oplus M$), CF03 (A8 hierarchical interfaces and void-debt throttling), CF04 (telegraph/Cattaneo causality), and CF10 (lattice $\rightarrow$ continuum program) [Lietz, 2026a,b,c,d].

2 Corrections versus CF11 v1.0

CF11 is a corrected rewrite intended to be CFN-ready (the notebook must implement the gates without inventing new math). The corrections are operational:

1. **Ad hoc projectors removed.** Invalid expressions such as “ $P_J = J/\sqrt{J^2 + M^2}$ ” are removed. The state-level split is now defined using spectral (Riesz) projections of a linearized generator (Section 5.2) [Kato, 1995].
2. **Orthogonality scoped correctly.** No global L^2 -orthogonality $\int \Phi_J \Phi_M = 0$ is asserted. When orthogonality is used, it is defined with respect to a declared quadratic form (energy Hessian / quantum metric) and only within the quadratic regime (Section 5.3).
3. **Stress–energy formulas corrected.** Curved-space Noether claims and non-minimal coupling mistakes are removed. The main text uses a minimal scalar-field proxy stress–energy only as a *derived-limit diagnostic* (Section 6) [Weinberg, 2008, Mukhanov, 2005].

4. **EoS limits corrected.** The w_J expression is corrected and placed as a bounded statement with explicit small parameters; the $w_M \rightarrow 0$ limit is derived via either non-relativistic defect kinematics or rapid oscillations in a quadratic minimum (Section 7) [Turner, 1983].
5. **Finite-speed transport aligned with CF04/CF03.** Telegraph speed $c = \sqrt{D/\tau}$ is treated as the causality cone speed, and hierarchy can throttle to $c_{\text{eff}} \leq c_0$ via void debt [Lietz, 2026b,c, Cattaneo, 1958, Vernotte, 1958].

3 Imports from earlier CF modules

3.1 Metriplectic structure

CF11 assumes the axiom-core metriplectic form (A4), expressed abstractly as

$$\partial_t q = J(q) \frac{\delta \mathcal{I}}{\delta q} + M(q) \frac{\delta \Sigma}{\delta q}, \quad J^\top = -J, \quad M^\top = M \succeq 0, \quad (1)$$

with degeneracy conditions

$$J(q) \frac{\delta \Sigma}{\delta q} = 0, \quad M(q) \frac{\delta \mathcal{I}}{\delta q} = 0, \quad (2)$$

and entropy production

$$\frac{d\Sigma}{dt} = \left\langle \frac{\delta \Sigma}{\delta q}, M(q) \frac{\delta \Sigma}{\delta q} \right\rangle \geq 0. \quad (3)$$

These are treated as measurable instrument gates in the VDM canon [Lietz, 2025a,b, 2026a].

3.2 Hierarchy throttling

CF03 provides a hierarchy program and a phenomenological speed throttling law of the form

$$c_{\text{eff}}(k) = c_0 e^{-\beta D_{\text{void}}(k)/2}, \quad D_{\text{void}}(k) \sim k \text{ for depth } k, \quad (4)$$

treated here as a derived-limit closure whose validity must be gated in any simulation using it [Lietz, 2026b].

3.3 Finite-speed transport

CF04 derives telegraph/Cattaneo transport and the cone speed relation

$$c = \sqrt{D/\tau}, \quad (5)$$

with numerical gates for light-cone preservation and dispersion consistency [Lietz, 2026c, Cattaneo, 1958, Vernotte, 1958, Goldstein, 1951].

4 A derived-limit proxy model for bookkeeping

Cosmology-facing statements require a concrete proxy that maps a VDM state q to (i) an energy-like conserved functional $\mathcal{I}$, (ii) an entropy Σ , and (iii) lattice-measurable local densities. CF11 uses a minimal scalar order parameter model with a metriplectic “thermostat” degree of freedom that enforces the degeneracy constraint $M \delta \mathcal{I} = 0$.

4.1 State and functionals

A naive “damped” scalar model on (Φ, Π) would typically drain $\mathcal{I}[\Phi, \Pi]$, violating (2). The metriplectic fix is to include a reservoir coordinate that stores removed mechanical energy while increasing entropy [Morrison, 1986, Öttinger, 2005].

Define the state

$$q = (\Phi, \Pi, u), \quad (6)$$

where $u(x, t)$ is an internal-energy (bath) density.

Choose a conserved functional

$$\mathcal{I}[\Phi, \Pi, u] = \int d^3x \left(\frac{1}{2} \Pi^2 + \frac{1}{2} |\nabla \Phi|^2 + V(\Phi) + u \right), \quad (7)$$

and an entropy functional

$$\Sigma[\Phi, \Pi, u] = \int d^3x s(u), \quad s'(u) > 0. \quad (8)$$

4.2 Concrete J and M satisfying degeneracy

Let J act canonically on (Φ, Π) and leave u untouched:

$$J = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (9)$$

Define M as a rank-1 positive semidefinite block coupling Π and u :

$$M(q) = \frac{1}{\tau s'(u)} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -\Pi \\ 0 & -\Pi & \Pi^2 \end{pmatrix} = \frac{1}{\tau s'(u)} v v^\top, \quad v = (0, 1, -\Pi). \quad (10)$$

Lemma 1 (Degeneracy and monotonicity in the proxy model). *With $\mathcal{I}$ and Σ defined in (7)–(8) and J, M as in (9)–(10), the degeneracy conditions (2) hold, $\mathcal{I}$ is conserved by the full flow, and Σ is non-decreasing.*

Proof. Compute variational derivatives: $\delta\Sigma/\delta q = (0, 0, s'(u))$ is supported only on u , and J does not move u , so $J \delta\Sigma = 0$. Also $\delta\mathcal{I}/\delta q = (-\nabla^2\Phi + V'(\Phi), \Pi, 1)$. Since $M \propto v v^\top$ with $v^\top(\delta\mathcal{I}/\delta q) = \Pi - \Pi \cdot 1 = 0$, one has $M \delta\mathcal{I} = 0$. Finally, $\dot{\Sigma} = \langle \delta\Sigma, M \delta\Sigma \rangle = \int (\Pi^2/\tau) d^3x \geq 0$. $\square$

4.3 Resulting evolution and reduced damped equation

The proxy evolution reads

$$\dot{\Phi} = \Pi, \quad (11)$$

$$\dot{\Pi} = \nabla^2\Phi - V'(\Phi) - \frac{1}{\tau}\Pi, \quad (12)$$

$$\dot{u} = \frac{1}{\tau}\Pi^2. \quad (13)$$

Eliminating Π yields the reduced damped Klein–Gordon/Allen–Cahn type equation

$$\ddot{\Phi} + \frac{1}{\tau}\dot{\Phi} - \nabla^2\Phi + V'(\Phi) = 0, \quad (14)$$

now explicitly understood as the projection of a fully metriplectic, energy-conserving system. CF11 uses this proxy strictly as a *bookkeeping model* to define measurable diagnostics and gates; it is not claimed as fundamental cosmological microphysics.

5 What “ J -sector” and “ M -sector” mean

5.1 Generator split versus state split

The metriplectic split is first a split of the *generator of motion*:

$$\dot{q} = \dot{q}_J + \dot{q}_M, \quad \dot{q}_J := J \frac{\delta\mathcal{I}}{\delta q}, \quad \dot{q}_M := M \frac{\delta\Sigma}{\delta q}. \quad (15)$$

It is *not* automatically a split of the state $q = q_J + q_M$. A state-level split requires an additional declared projection choice. CF11 fixes that choice and makes it testable.

5.2 Mode-projection definition via spectral (Riesz) projectors

Let $\bar{q}(t)$ be a coarse-grained background (e.g., spatial mean on a comoving window ℓ). Let perturbations be $\delta q := q - \bar{q}$ and linearize the dynamics:

$$\partial_t(\delta q) = \mathcal{L}(t) \delta q + \mathcal{N}(\delta q), \quad (16)$$

where $\mathcal{L}$ is the linearized generator and $\mathcal{N}(\delta q) = \mathcal{O}(\|\delta q\|^2)$ collects nonlinear terms.

Definition 1 (Spectral-gap assumption (S-gap)). On a declared time window and coarse-graining scale, the spectrum of $\mathcal{L}(t)$ separates into: (i) an oscillatory band with $|\Re \lambda| \ll |\Im \lambda|$; and (ii) a relaxational band with $\Re \lambda \leq -\gamma < 0$, for some γ that is bounded away from zero over the window.

Under (S-gap), define Riesz projectors P_J and P_M onto the two spectral sets by a contour integral representation, and set

$$\delta q = \delta q_J + \delta q_M, \quad \delta q_J := P_J \delta q, \quad \delta q_M := P_M \delta q, \quad P_J P_M = 0. \quad (17)$$

This is the mathematically standard replacement for ad hoc “operator algebra” between J and M [Kato, 1995]. The assumption itself is not philosophical: it is a T2 simulation gate (Section 11).

5.3 Sector energies and cross-terms

For the proxy model, the quadratic energy in perturbations around $\bar{q}$ is

$$\delta^2 \mathcal{I}[\delta q] = \frac{1}{2} \langle \delta q, H_{\mathcal{I}}(\bar{q}) \delta q \rangle, \quad (18)$$

where $H_{\mathcal{I}}$ is the Hessian (second variation) of $\mathcal{I}$ at $\bar{q}$. Define sector energies

$$E_J := \frac{1}{2} \langle \delta q_J, H_{\mathcal{I}} \delta q_J \rangle, \quad E_M := \frac{1}{2} \langle \delta q_M, H_{\mathcal{I}} \delta q_M \rangle, \quad (19)$$

and the quadratic interaction remainder

$$E_{\times} := \langle \delta q_J, H_{\mathcal{I}} \delta q_M \rangle. \quad (20)$$

If P_J, P_M are chosen to be orthogonal with respect to the $H_{\mathcal{I}}$ -inner product (a standard modal-analysis choice), then $E_{\times} = 0$ at quadratic order. Beyond the quadratic regime, the remainder is controlled by nonlinearities and must be treated as a measured error term, not assumed away.

6 Effective stress–energy and two-fluid bookkeeping

6.1 Scalar-field proxy stress–energy in FRW

In an FRW background with scale factor $a(t)$, a minimally coupled scalar has effective energy density and pressure (spatially averaged under isotropy assumptions)

$$\rho[\Phi] = \frac{1}{2} \dot{\Phi}^2 + \frac{1}{2a^2} |\nabla \Phi|^2 + V(\Phi), \quad (21)$$

$$p[\Phi] = \frac{1}{2} \dot{\Phi}^2 - \frac{1}{6a^2} |\nabla \Phi|^2 - V(\Phi). \quad (22)$$

These are used only as derived-limit diagnostics for lattice analogues [Weinberg, 2008, Mukhanov, 2005].

6.2 Sector densities, pressures, and tracked remainders

Define coarse-grained sector densities and pressures by reconstructing sector fields from $\delta q_J, \delta q_M$ and inserting into the quadratic approximations of ρ and p , then averaging over a comoving window of size ℓ :

$$\rho_J := \langle \rho[\Phi_J] \rangle_{\ell}, \quad p_J := \langle p[\Phi_J] \rangle_{\ell}, \quad (23)$$

$$\rho_M := \langle \rho[\Phi_M] \rangle_{\ell}, \quad p_M := \langle p[\Phi_M] \rangle_{\ell}, \quad (24)$$

with explicit tracked remainders

$$\rho_{\times} := \langle \rho[\Phi] - \rho[\Phi_J] - \rho[\Phi_M] \rangle_{\ell}, \quad (25)$$

$$p_{\times} := \langle p[\Phi] - p[\Phi_J] - p[\Phi_M] \rangle_{\ell}. \quad (26)$$

No smallness of $(\rho_{\times}, p_{\times})$ is assumed; instead, their magnitude is gated (Section 11).

7 Equations of state: controlled limits and correct bounds

7.1 J -sector: vacuum-like limit $w_J \approx -1$

Define

$$K_J := \frac{1}{2} \langle \dot{\Phi}_J^2 \rangle_\ell, \quad G_J := \frac{1}{2a^2} \langle |\nabla \Phi_J|^2 \rangle_\ell, \quad U_J := \langle V(\Phi_J) \rangle_\ell, \quad (27)$$

so $\rho_J = K_J + G_J + U_J$ and $p_J = K_J - \frac{1}{3}G_J - U_J$. Introduce dimensionless small parameters

$$\epsilon_J := \frac{K_J}{U_J}, \quad \delta_J := \frac{G_J}{U_J}. \quad (28)$$

Then the EoS satisfies the exact identity

$$w_J = \frac{\epsilon_J - \frac{1}{3}\delta_J - 1}{\epsilon_J + \delta_J + 1}. \quad (29)$$

In particular, if $\epsilon_J \leq \epsilon_{\max}$ and $\delta_J \leq \delta_{\max}$ with $\epsilon_{\max} + \delta_{\max} < 1$, then

$$|1 + w_J| \leq \frac{2\epsilon_{\max} + \frac{2}{3}\delta_{\max}}{1 - \epsilon_{\max} - \delta_{\max}}. \quad (30)$$

Thus $w_J \approx -1$ becomes a *measured* conclusion only when diagnostics enforce $\epsilon_J, \delta_J \ll 1$ [Mukhanov, 2005, Weinberg, 2008].

7.2 M -sector: dust-like limit $w_M \approx 0$

Two distinct, testable routes yield $w_M \approx 0$. CF11 permits either (or both); lattice measurements decide which regime is realized.

Route A: non-relativistic defect/particle kinematics. If the M -sector coarse-grains to localized structures with RMS velocity dispersion $\sigma_v \ll 1$, then the pressure is kinetic and suppressed by v^2 , yielding

$$w_M \approx \frac{\sigma_v^2}{3}. \quad (31)$$

Lattice proxy: identify defect cores/interface segments (CF03-style) and track their velocities to estimate σ_v [Lietz, 2026b].

Route B: rapid oscillations near a quadratic minimum. Near a stable minimum Φ_* , any smooth potential is locally quadratic:

$$V(\Phi) \approx \frac{1}{2}m^2(\Phi - \Phi_*)^2. \quad (32)$$

If the M -sector is dominated by coherent oscillations about Φ_* with $m \gg H$ and negligible gradient energy, then time-averaging over oscillations gives $\langle K_M \rangle \approx \langle U_M \rangle$, hence $\langle w_M \rangle \approx 0$ [Turner, 1983, Mukhanov, 2005].

8 Energy exchange Q^μ and entropy constraints

8.1 Bookkeeping definition

Given a split $T^{\mu\nu} = T_J^{\mu\nu} + T_M^{\mu\nu} + T_\times^{\mu\nu}$ with tracked remainder, define the exchange current

$$Q^\nu := -\nabla_\mu T_J^{\mu\nu}. \quad (33)$$

Then

$$\nabla_\mu (T_J^{\mu\nu} + T_M^{\mu\nu}) = -\nabla_\mu T_\times^{\mu\nu}. \quad (34)$$

In the controlled regime where $T_\times$ is small by gate, the effective two-fluid description is consistent:

$$\nabla_\mu T_J^{\mu\nu} \approx -Q^\nu, \quad \nabla_\mu T_M^{\mu\nu} \approx +Q^\nu. \quad (35)$$

Interacting-fluid phenomenology is standard, but CF11 treats it strictly as an *emergent bookkeeping* after the spectral split, not as a new fundamental coupling postulate [Amendola, 2000, Wang et al., 2016].

8.2 Constraint from the entropy law

The axiom-core entropy law (3) imposes a sign constraint on irreversible transfers. In the proxy model (11)–(13), $\dot{\Sigma} = \int (\Pi^2/\tau) d^3x \geq 0$ implies that kinetic damping is non-negative (up to reversible transfers). Operationally, any inferred Q^μ must be compatible with $\dot{\Sigma} \geq 0$ under the sign conventions used in the analysis.

8.3 Coincidence and model selection are closure problems

A metriplectic system guarantees monotone relaxation of an appropriate Lyapunov functional, but it does *not* uniquely determine a cosmological $Q(z)$ without a closure linking coarse observables to $(J, M, \mathcal{I}, \Sigma)$. CF11 therefore treats “coincidence attractors” as pre-registrable hypotheses only after a specific closure is declared; that closure is then judged by out-of-sample prediction and model selection metrics such as AIC/BIC [Akaike, 1974, Schwarz, 1978, Burnham and Anderson, 2002].

9 Causality and stability: what is and is not guaranteed

9.1 Finite-speed propagation

Telegraph/Cattaneo transport implies a finite front speed $c = \sqrt{D/\tau}$ [Cattaneo, 1958, Vernotte, 1958, Goldstein, 1951]. Hierarchical interfaces can throttle effective speed via (4) [Lietz, 2026b,c]. In simulations, measured signal fronts must satisfy $|x| \leq c_{\text{eff}}t$ within declared tolerance (Section 11).

9.2 Stability is conditional; tachyonic phases are allowed

CF03 intentionally uses tachyonic/unstable regimes to generate interfaces. CF11 therefore makes no claim of global linear stability. The conditional contract is: (i) in post-condensation regions where the local curvature $V''(\Phi_*) > 0$, linearized perturbations of the proxy model are damped by τ^{-1} ; and (ii) instability in regions where $V'' < 0$ is a feature (interface formation) and must satisfy the CF03 gate suite, not be suppressed.

10 A8 hierarchy connection and a minimal lensing observable

CF11 connects hierarchy to cosmology only operationally. CF03 predicts hierarchical interface formation and “void debt” accumulation, which modulates transport speeds and propagation-based observables. CF11 packages this as a minimal, falsifiable correlation statistic rather than a hard-coded density law.

10.1 Hierarchy–lensing bias statistic

Define

$$\beta_{\text{bias}} := \frac{\kappa_{\text{obs}} - \kappa_{\text{baseline}}}{\kappa_{\text{baseline}}}, \quad (36)$$

where κ_{baseline} is the convergence predicted by a declared baseline model (e.g., $\Lambda\text{CDM} + \text{NFW}$), and κ_{obs} is measured from data [Bartelmann and Schneider, 2001]. The qualitative prediction is that β_{bias} correlates with a hierarchy indicator (interface density, void debt proxy, or other CF03-derived statistic). This becomes a binary claim only when a specific indicator and dataset are pre-registered (Section 11).

11 Falsification gates (binary, pre-registrable)

CF11 is designed to be killed quickly. Gates are grouped by tier: internal T2 simulation gates and external T3+ observational regression gates. Thresholds shown here are defaults and must be fixed (with dataset and code gates) before a run.

11.1 T2 internal gates (lattice / simulation)

Gate: G-CF11-T2-1 (Degeneracy residuals) *Threshold:* $\|J \delta \Sigma\| / \|\delta \Sigma\| \leq 10^{-12}$ and $\|M \delta \mathcal{I}\| / \|\delta \mathcal{I}\| \leq 10^{-12}$ (relative), measured per checkpoint.

Verify the metriplectic degeneracy conditions (2) within declared numerical tolerance. This is an instrument validity gate [Lietz, 2025b, 2026a].

Gate: G-CF11-T2-2 (Entropy law) *Threshold: $\Delta\Sigma \geq -\varepsilon_\Sigma$ per step; default $\varepsilon_\Sigma \sim 10^{-15}$ (fp64).*

Verify the discrete entropy monotonicity consistent with (3). Negative excursions beyond rounding tolerance fail the gate.

Gate: G-CF11-T2-3 (Spectral-gap validity of the mode split) *Threshold: exists $\gamma > 0$ such that all “relaxational” modes satisfy $\Re\lambda \leq -\gamma$ over the declared window.*

Compute $\text{spec}(\mathcal{L}(t))$ (or the Jacobian of the discretized update) and verify the separation in Definition 1. This gate controls whether (17) is meaningful [Kato, 1995].

Gate: G-CF11-T2-4 (Cross-term control) *Threshold: $|\rho_\times|/(\rho_J + \rho_M) \leq \varepsilon_\times$ after coarse-graining; default $\varepsilon_\times = 0.05$.*

Compute the tracked remainder $\rho_\times$ and require that the split is not dominated by uncontrolled interactions in the intended regime.

Gate: G-CF11-T2-5 (EoS regime discipline) *Threshold: apply w_J and w_M band tests only if corresponding diagnostics pass.*

For $w_J \approx -1$, require $\epsilon_J, \delta_J \ll 1$ (e.g., $\epsilon_J, \delta_J \leq 0.05$). For $w_M \approx 0$, require either $\sigma_v \ll 1$ or $m/H \gg 1$ and $\langle K_M \rangle / \langle U_M \rangle \rightarrow 1$. This prevents circular reasoning.

Gate: G-CF11-T2-6 (Causality / cone bound) *Threshold: measured front speed $v_{\text{front}} \leq c_{\text{eff}}(1 + \eta)$; default $\eta = 0.02$.*

Signals must respect the telegraph cone speed (possibly throttled by hierarchy) within declared tolerance [Lietz, 2026c].

11.2 T3+ observational gates (cosmology regression)

Gate: G-CF11-T3-1 (Hierarchy lensing bias) *Threshold: either detection: $|\beta_{\text{bias}}| > 5\sigma$, or null: $|\beta_{\text{bias}}| < 1\sigma$ after controls.*

On a pre-registered dataset, test the correlation between β_{bias} and a hierarchy indicator. A robust null falsifies the hierarchy→lensing channel in CF11.

Gate: G-CF11-T3-2 (Model selection) *Threshold: pre-registered ΔAIC or ΔBIC threshold.*

On a declared probe set (e.g., BAO+SNe+CMB distance priors + lensing), require a pre-registered model selection outcome. Typical interpretations follow [Burnham and Anderson, 2002] (e.g., $\Delta\text{AIC} \leq -2$ “VDM preferred”; $\Delta\text{AIC} \geq +10$ “VDM rejected”).

Gate: G-CF11-T3-3 (Interacting-fluid coupling closure) *Threshold: if a specific $Q(z; \theta)$ is pre-registered: either $\theta > 3\sigma$ with $\hat{\Sigma} \geq 0$ consistency, or $|\theta| < 1\sigma$.*

This gate treats interacting-fluid couplings as empirical parameters constrained jointly by data and the entropy-sign discipline.

12 Implementation notes: computing the split on a lattice

At each checkpoint time t_i , the mode split (17) can be computed without inventing new math:

1. Linearize the discretized update rule to obtain the Jacobian L_i approximating $\mathcal{L}(t_i)$.

2. Compute a real Schur decomposition $L_i = QTQ^\top$ [Golub and Van Loan, 2013].
3. Identify eigenvalues belonging to the oscillatory vs relaxational bands using a declared criterion (e.g., $\Re\lambda \leq -\gamma$).
4. Build projectors P_J, P_M from Schur blocks (invariant subspace extraction).
5. Project δq to obtain $\delta q_J, \delta q_M$, then compute ρ_J, ρ_M, w_J, w_M and the remainders $(\rho_\times, p_\times)$.

The projection criterion (γ , band definitions, coarse-graining scale ℓ) must be declared up front and is constrained by gates G-CF11-T2-3 and G-CF11-T2-4.

13 Canon integration (proposed new entries)

This CF proposes registering the following items in the equations and symbols canons (numbers provisional; final anchors are assigned by the registry):

- **VDM-E-145 (Mode split via spectral projectors):** $\delta q = P_J \delta q + P_M \delta q$, $P_J P_M = 0$.
- **VDM-E-146 (Sector energies from Hessian):** (19).
- **VDM-E-147 (J-sector EoS identity):** (29) and bound (30).
- **VDM-E-148 (Dust limits):** (31) and the rapid-oscillation average $\langle w_M \rangle \approx 0$.
- **VDM-E-149 (Exchange current definition):** (33).

Proposed new symbols. P_J, P_M (spectral projectors); γ (gap threshold); ℓ (coarse-graining scale); $\rho_\times, p_\times$ (interaction remainders); ϵ_J, δ_J (diagnostic ratios); σ_v (velocity dispersion proxy); β_{bias} (hierarchy–lensing bias statistic).

14 Red-team checklist

CF11 is designed to withstand the most likely criticisms:

1. **“This is just relabeling Λ CDM.”** The split is not a label: it is a measurable decomposition (spectral gap + cross-term control) with gates and contradiction reports.
2. **“The split is arbitrary.”** The only arbitrariness is the coarse-graining scale ℓ and band cut γ , both declared and constrained by the spectral-gap and remainder gates.
3. **“You assume $w \approx 0$ and $w \approx -1$.”** CF11 provides conditions under which those limits follow and requires regime diagnostics ($\epsilon_J, \delta_J, m/H, \sigma_v$) before applying EoS band tests.
4. **“Energy exchange violates conservation.”** Total conservation holds at the axiom-core level; Q^μ is internal bookkeeping after the split and is additionally constrained by $\dot{\Sigma} \geq 0$.

15 Closure statement

CF11 reduces the “dark sector from metriplectic dynamics” idea to a killable contract: (i) a defined, implementable spectral split (Riesz projectors) with measurable spectral-gap validity; (ii) sector stress–energy with tracked remainders rather than hand-waved cancellations; (iii) bounded EoS limits that become binary only under explicit regime diagnostics; and (iv) a two-level gate suite (T2 lattice, T3+ observational regression) that yields contradiction reports on failure.

Tier. T1 complete (formalism + gates). The next step is T2 lattice validation implementing the spectral split and remainders, followed by T3 observational regression.

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